COSC 360L	Networking Lab	1(0,2)
This Lab helps prepare students seeking to pass advanced Cisco Certifications. The student will acquire knowledge of network equipment and protocol functionality, how to build a simple Ethernet network using routers, switches, and computers, and how to use router CLI commands to perform basic configuration and verification. The student will also learn valuable network problem-solving techniques and concepts. **Co-requisite**: COSC 360 or CCEE 354. Equivalent to CCEE 354L.		

COSC 399	Co-op Training Experience	1(1,0)
Each student must complete 8 weeks of practical training in a field related to their interests. This Co-op work experience is usually fulfilled during the summer semester of the third year in the program. Students must submit a formal report and poster and make a formal presentation about their Co-op experience. **Prerequisite***:* ENGL 217; Senior Standing.		

COSC 421	Theory of Computation	3(3,0)
This course covers theoretical principles embodied in automata and grammars. Topics include Deterministic, Non-deterministic, and Epsilon Non-deterministic Finite Automata (DFA, NFA, and e-NFA), DFA minimization, equivalence of NFAs and DFAs, Moore and Mealy machines, closure properties, regular expressions, regular languages, regular grammar, pumping lemma, context-free languages, context-free grammar, pushdown automata, Turing machines, and other selected topics as time permits. **Prerequisites***:* COSC 215 and MATH 210		

COSC 451	Operating Systems	3(3,0)
Students learn the concepts of operating systems theory and their implementations. The course covers computer and operating system structures, process and thread management, process synchronization, deadlocks, security and protection, inter-process communication, memory management, and I/O control systems. A case study of a UNIX operating system will accompany the course. **Prerequisite***:* COSC 214. Equivalent to CCEE 513.		

COSC 490	Summative Learning Experience	3(3,0)
This is a yearlong graduation project focused on individual problems and research. Each student must complete it in their emphasis area. **Prerequisite***:* ENGL 217.		

Business Required Courses

As described below, the COSC program requires two business courses, BMKA 200 and BMGT 300.

BMGT 300	Project Management	3(3,0)
This course provides students with the skills to manage their business projects using effective techniques for leading, organizing, scheduling, and controlling tasks that advance project goals. Topics include project selection and statement of work; project managers' skills and task breakdown structure; PERT/CPM scheduling and budgeting. **Prerequisite:** Junior Standing.		

BMKA 200	Introduction to Marketing	3(3,0)
This course introduces the basic principles, theories, and marketing practices in our modern, ever-changing business environment. The course covers the marketing process activities of creating value for customers and capturing value from customers in return. It also discusses the marketing mix and how to build long-term customer relationships. Students will analyze case studies about a “real-life” product or service. Videos and in-class discussions on current marketing topics will assist in the learning experience. **Co-requisite**: ENGL 210		

II. Elective Courses

Descriptions of the major elective courses are given below.

CCEE 426	Design of Embedded Systems	3(3,0)
This course addresses the design of embedded real-time systems, models of computation, validation techniques, and automatic synthesis. Finite-state machines, synchronous languages, data-flow networks, Petri nets, software optimization and performance estimation, operating systems and scheduling, system-level simulation, and interface-based design. **Prerequisite**: CCEE 221. Equivalent to COSC 356.		

COSC 434	Advanced Web Programming	3(3,0)
This course builds on the web programming course and prepares students to become full-stack developers. It allows students to learn how to develop back-end programs, connect their website or web application to a database, use regular expressions, implement asynchronous client access to server data, work with structured data formats such as JSON, and work with third-party media like web services. **Prerequisites:** COSC 231 and COSC 333. Equivalent to CCEE 514.		

COSC 435	Mobile Application Development	3(2,2)
This course examines the principles of mobile application design, development, and testing. It covers memory management, user interface design, input methods, data handling, and network techniques. Students are expected to work on a project that produces a professional-quality mobile application. Projects will be deployed in real-world applications. **Prerequisite***:* COSC 214, Equivalent to CCEE 511		

COSC 436	Game Programming	3(2,2)
This course provides an overview of software technologies for game development. It gives students a conceptual understanding of game design and practical experience in game creation. **Prerequisites***:* COSC 214; Senior standing.		

COSC 438	Introduction to Software Testing	3(3,0)
This course will introduce students to software testing, its importance, goals, approaches, and the roles of major players (developers, testers, and users). Students will learn about the software testing life cycle, standards, types of testing (manual vs. automated), testing methods (black-box vs. white-box), testing levels (functional, unit		

testing…), and documentation. This course will also cover hands-on experience with automated testing tools. **Prerequisite***:* COSC 341

COSC 460L	**Networking Lab 2**	**1(0,2)**
This Lab helps prepare students seeking to pass advanced Cisco Certifications (CCNA 1). The student will acquire knowledge of the functionalities of network equipment and protocols, how to build a simple Ethernet network using routers, switches, and computers, and how to use router CLI commands to perform basic configuration and verification. The student will also learn valuable network problem-solving techniques and concepts through hands-on skills using hardware devices. **Co-requisite**: COSC 360L or CCEE 354L. Equivalent to CCEE 554L.		

COSC 461	**Advanced Networking**	**3(3,0)**
This course prepares students to act as System and Network Administrators by implementing Active Directory Domain Services (AD DS) in distributed environments that can include complex network services and domain controllers. The covered materials help students efficiently automate the administration of users, groups, and computers. **Prerequisite***:* COSC 360 or CCEE 354*.*		

COSC 461L	**Networking Lab 3**	**1(0,2)**
This Lab helps prepare students seeking to pass the Cisco Routing and Switching Essentials (CCNA 2) Certification. The primary focus of this Lab is routers and switches in small networks. The student will learn the architecture, components, and operations of routers and switches, as well as their basic configuration and functionality. They will also learn how to configure and troubleshoot static and dynamic routing protocols, access-control lists, VLANs, DHCP for IPv4 and IPv6 networks, and NAT operations. **Pre-requisite**: COSC 460L or CCEE 554L. Equivalent to CCEE 555L.		

COSC 462L	**Networking Lab 4**	**1(0,2)**
This lab helps prepare students seeking to pass the Cisco Routing and Switching Essentials (CCNA 3) Certification. The primary focus of this Lab is routers and switches in large and complex networks. The student will learn how to configure routers and switches with advanced functionalities. They will also learn how to configure and troubleshoot enhanced switching technologies, first-hop redundancy protocols in a switched network, wireless routers and clients, and routers in complex networks. Finally, they will learn to manage CISCO IOS software licensing and configuration files. **Co-requisite**: COSC 461L or CCEE 555L. Equivalent to CCEE 556L.		

COSC 463	**Cybersecurity operations**	**3(3,0)**
This hands-on, career-oriented course focuses on practical topics to prepare students to acquire the skills required to work in the cybersecurity domain. The course will help students seeking to pass the Cisco Cybersecurity Operations Associate certification. **Pre-requisite**: COSC 360 (or CCEE 354).		

COSC 480	Machine Learning and Data Mining	3(3,0)
This course introduces students to the basic knowledge representation and learning techniques. The emphasis is on understanding the data mining process, applying appropriate approaches to solve practical problems, and developing intelligent software applications. The course covers several topics about classification, prediction, and clustering. **Prerequisites***:* COSC 214 and MATH 351 or BADM 250. Equivalent to CCEE 564.		

COSC 481	Artificial Intelligence	3(3,0)
This course introduces students to the fundamentals of knowledge representation and learning in artificial intelligence. The emphasis will be on understanding fundamental concepts in artificial intelligence and applying corresponding approaches to solve practical problems and develop useful software applications. Covered topics include Intelligent agents, informed and uninformed search strategies, and adversarial search. The Python language libraries will also be introduced. **Prerequisite**: COSC 214. Equivalent to CCEE 562.		

COSC 482	Data Science and Web Scraping	3(3,0)
Data is becoming the fuel of the 21st century, and acquiring data processing and analysis skills is becoming necessary. This course introduces data science processes, with a focus on web scraping as an application. The course will combine different domains, i.e., web programming, system programming, and machine learning. In particular, the course focuses on analyzing webpage HTML using Python, extracting available information, and generating dashboards. Co-requisite: COSC 333 (or CCEE 411) and MATH 351 (or BADM 250).		

COSC 496	Independent Study I	3(3,0)
This course allows students to explore new academic research/study that is unavailable in regular courses. It might be an interdisciplinary research project in a field of special interest. **Prerequisites**: Senior Standing and a CGPA of 80 and above.		

COSC 497	Independent Study II	3(3,0)
This course allows students to explore new academic research/study that is unavailable in regular courses. It might be an interdisciplinary research project in a field of special interest. **Prerequisites**: Senior Standing and a CGPA of 80 and above.		

COSC 498	Special Topics in Computer Science	1-3 credits
A special topic course to be defined based on current trends and departmental special interests. It may incorporate technical seminars, advanced readings, and special projects as required/approved by the department. **Prerequisite**: Senior standing.		